# A One-Sided Level-Crossing Budget for Physics-Informed Neural Networks:
# Validated Mechanism, Absent Pathology

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**Abstract**

Physics-informed neural networks (PINNs) on stiff and unstable PDEs are widely reported to fail through runaway high-frequency oscillation—most pointedly on the complex Ginzburg–Landau equation (CGLE), where a recent architecture-level fix (ASPEN) is premised on a conventional MLP PINN "catastrophically diverging into non-physical oscillations." We design the natural *loss-level* counterpart: a one-sided *level-crossing budget*, built on the Kac–Rice/co-area machinery of our companion work, that penalizes the network field's spatial crossing density only where it exceeds the physical budget of the solution class. The loss is mesh-free, FFT-free, evaluated at the collocation points a PINN already uses, costs one spatial derivative the residual already computes, and is provably inactive—zero loss, zero gradient—on fields within budget. We validate the mechanism in vitro: on sparse-data interpolation where a periodic-activation network demonstrably over-oscillates, the budget clamps the crossing profile to specification and improves test error on every seed. We then report, with full receipts, a negative result in vivo: across a faithful reimplementation of the ASPEN benchmark (two input conventions, up to the full published training depth) and a Benjamin–Feir-unstable chaotic testbed, the vanilla PINN always fails—but always by *under*-resolving or decorrelating, never by excess oscillation. Its crossing density sits below the physical budget at every level, so a one-sided budget is verifiably inert, and uniform gradient damping is inert or harmful for the same reason. We conclude that loss-level oscillation suppressors—ours included—address a failure mode whose prevalence is implementation-sensitive, and we release a crossing-profile diagnostic that distinguishes over- from under-oscillatory PINN failure in one plot, together with all code, reference solvers, and raw runs.

## 1 Introduction

Physics-informed neural networks [11] minimize a PDE residual at collocation points instead of fitting data on a grid. Their failure modes are by now a literature of their own [10, 15]: stiff dynamics, chaotic regimes, and long horizons produce solutions that are smooth, plausible, and wrong—or, in the telling of several recent papers, fields that *blow up in high frequency*. On the complex Ginzburg–Landau equation (CGLE), the canonical stiff nonlinear testbed [2], the ASPEN architecture [3] is motivated by exactly this claim: a conventional MLP PINN "develops spurious, high-frequency oscillations that grow in time, rendering the prediction physically meaningless."

If the pathology is excess oscillation, there is a natural *loss-level*, architecture-agnostic counterpart to spectral architecture fixes. In companion work on implicit neural representations [6] we turned the Rice level-crossing density [7, 12]—the number of times a field crosses a level per unit length, a closed-form spatial proxy for frequency content [9]—into a differentiable training signal via the co-area formula and a smoothed Monte-Carlo estimator. There, the loss *demands* crossings that spectrally biased networks under-produce. Here we flip its sign: a **one-sided crossing budget** that penalizes the field only where its crossing density *exceeds* what the physics allows. One mechanism, both directions.

The budget has three properties tailored to PINNs. It is evaluated pointwise at the scattered collocation points themselves—no grid, no FFT, exactly where frequency-domain penalties are awkward. It reuses the spatial derivative the residual loss already computes, so its marginal cost is one kernel evaluation. And it is *one-sided by construction*: on a field within budget it contributes zero loss and, provably, zero gradient—it cannot push a smooth field toward oscillation, the failure it is designed to prevent. Budgets come from three sources of decreasing strength: a reference spectral simulation, a closed-form physics prior (for a monotone-front class, each level is crossed once per front), or a single scalar cap.

**Pre-registered honesty.** This study was gated and kill-conditioned in advance: the mechanism had to demonstrably steer crossing profiles (established in the companion work and re-verified in vitro here), the vanilla failure had to reproduce, and—the condition that fired—the failure had to actually *be* the excess-oscillation pathology. We report what happened, in full.

**Contributions.** (1) *The loss*: a validated, one-sided, physically calibrated crossing-density budget with three budget sources and a zero-gradient-under-budget guarantee (Section 3). (2) *An in-vitro positive control*: on sparse-data interpolation with a periodic-activation network—a setting that genuinely over-oscillates—the budget clamps the crossing profile to specification and improves test error on every seed (Section 4). (3) *A receipted negative result*: on a faithful reimplementation of the ASPEN CGLE benchmark (both raw and normalized inputs, 30k and the full published 100k iterations) and on a Benjamin–Feir-unstable chaotic CGLE testbed, vanilla PINNs fail *under*-oscillatorily in every run; the budget—and uniform gradient damping—are verifiably inert on such failures (Section 5). (4) *A diagnostic*: the measured crossing profile against the physical budget separates over- from under-oscillatory failure in a single plot, and would have identified the mismatch between pathology and remedy before any stabilizer was trained (Section 3).

# 2 Related work

**PINN failure modes and schedules.** Krishnapriyan et al. [10] characterize why residual training fails on seemingly simple PDEs and propose curriculum/seq2seq regimes; Wang et al. [15] enforce temporal causality through residual weighting. Both address the *propagation* failure—early times fit, later times lost—which is precisely the mode we observe on the CGLE front benchmark. Architecture-level spectral remedies include Fourier features [14], ASPEN's adaptive spectral layer [3], and multi-scale SIREN variants for Ginzburg–Landau dynamics [5].

**Crossing statistics.** The Rice formula and its co-area generalization [1, 4, 7, 12] give the expected level-set measure of a field; zero-crossing rates as frequency surrogates are classical [9]. Our companion paper [6] develops the differentiable estimator and studies the *demand* direction for implicit neural representations; to our knowledge no prior work uses a crossing *budget* as an anti-oscillation regularizer for PINNs—and this paper's finding is that on the tested CGLE regimes, nothing does, because the pathology it targets did not occur.

# 3 The one-sided crossing budget

Let $f_\theta$ be one real component of the PINN field (we budget $\operatorname{Re} A$ and $\operatorname{Im} A$ separately). With the smoothed Monte-Carlo crossing density of the companion work,

$$\hat{c}_\varepsilon(u) \;=\; \frac{1}{N}\sum_{i=1}^{N} \delta_\varepsilon\big(f_\theta(x_i,t_i)-u\big)\,\big|\partial_x f_\theta(x_i,t_i)\big|, \tag{1}$$

evaluated at the PDE collocation points $(x_i, t_i)$ (Latin hypercube over space-time; $\partial_x f_\theta$ is shared with the residual computation), the budget loss at fixed levels $u_1, \ldots, u_L$ spanning the solution class's amplitude range is

$$\mathcal{L}_{\text{budget}} = \frac{1}{L}\sum_{j=1}^{L}\left[\frac{\operatorname{relu}\big(\hat{c}_\varepsilon(u_j)-b_j\big)}{b_j+\bar{b}}\right]^2, \qquad \bar{b} = \tfrac{1}{L}\textstyle\sum_j b_j. \tag{2}$$

Levels are *fixed a priori*—a PINN has no ground-truth field from which to take quantiles—and the one-sidedness is mandatory: early in training the field is smooth, and a symmetric matching loss would *pump oscillation in*, the exact failure being prevented. Under budget, (2) is identically zero on an open set of fields, so its gradient vanishes as well; our released unit tests verify both properties, along with the estimator's agreement with exact crossing counts and the Rice formula (within 5% and 10% respectively, reproduced from the companion work).

**Budget sources.** (a) *Reference simulation*: per-level crossing densities measured on a spectral/finite-difference reference of the same solution class (95th percentile over time, ×1.5 slack)—the strongest budget, but it presumes a solver, which we state plainly. (b) *Physics prior*: for the monotone-front class of the ASPEN benchmark, a front crosses each level in its range once per passage, so $b_j = m/L_{\mathrm{dom}}$ with a small allowance $m$ for shed transients—no simulation, no spectrum. For statistically stationary classes the Rice form $b \approx \frac{1}{\pi}\sqrt{\lambda_2/\lambda_0}$ from the physical spectrum plays the same role. (c) *Scalar cap*: one number for all levels.

**The profile as a diagnostic.** Independently of training, comparing the *measured* profile $\hat{c}_\varepsilon(u)$ of any trained field against $b_j$ classifies the failure mode: profile above budget ⇒ excess oscillation (the budget's patient); profile below budget with large error ⇒ under-resolution or decorrelation, for which oscillation suppressors of any kind are the wrong medicine. Every conclusion in Section 5 is legible in this one plot (Figure 1).

# 4 In vitro: the mechanism works where its pathology exists

Before the PDE setting, we verify the budget on a problem that demonstrably over-oscillates: a SIREN [13] ($\omega_0$=30, width 128, 3 hidden layers) fit to 40 random samples of a smooth signal $y = \tanh(3x) + 0.3\sin(2\pi x)$ on $[-1, 1]$. The physics-prior budget allows $m$=6 crossings per level over the domain ($b$=3 per unit length). Across three seeds, the MSE-only network exceeds the budget (peak crossing density 3.4–4.3) and interpolates with spurious oscillation; adding (2) at weight 1 clamps the peak profile to 2.6–2.7—strictly within budget—and *improves* dense-grid test MSE in every seed ($0.0233 \to 0.0198$, $0.0274 \to 0.0231$, $0.1099 \to 0.0649$). The mechanism does exactly what it claims when excess crossings exist (Figure 2).

# 5 In vivo: the pathology does not show up

## 5.1 Testbeds and protocol

**ASPEN front benchmark (faithful reimplementation).** CGLE $\partial_t A = A + (1+ib)\partial_{xx}A - (1+ic)|A|^2 A$ with $b$=0.5, $c$=$-1.3$, $x \in [-10, 7.5]$, $t \in [0, 10]$, $A(x, 0) = \tanh(-x)$, Dirichlet ends [3]: a stiff domain-wall relaxation. Vanilla PINN: 8×40 tanh MLP, Adam $10^{-3} \to 10^{-4}$, 20,000 residual + 1,000 IC + 1,000 BC Latin-hypercube points, $w_{\mathrm{res}}$=1, $w_{\mathrm{ic,bc}}$=100—ASPEN's published configuration. Reference: method-of-lines RK4 at their $\Delta t$=$10^{-4}$, $N_x$=1024, grid-convergence verified. We run both normalized-input and raw-coordinate variants, at 30k iterations (3 seeds) and at the full published 100k depth. [TODO: confirm 100k numbers when runs land]

**Benjamin–Feir-unstable testbed.** The front benchmark is BF-stable ($1 + bc = 0.35 > 0$); genuine broadband dynamics needs the unstable regime. We add periodic CGLE with $b$=2, $c$=$-1.2$ ($1 + bc = -1.4$: phase turbulence), $L$=64, ETDRK4 reference [8] run to the attractor, PINN horizon $T$=5 from an attractor state, periodicity hard-wired through Fourier coordinate embedding so no BC loss is needed.

**Metrics.** Relative $L_2$ against the reference on its space-time grid; radially banded spatial-spectrum error; the crossing profile of Section 3; temporal turning-point rates.

### 5.2 Vanilla failure, wrong disease

[TODO: Table with final numbers once exp4/diag100k land. Current verified content:] On the front benchmark the vanilla PINN fails reproducibly—relative $L_2$ $0.867 \pm 0.016$ over three seeds, physically useless—but the failure is *propagation loss*, not oscillation: error grows from 0.10 ($t<2$) to 0.76 ($t>5$), amplitudes stay bounded ($\max |A| = 1.05$), temporal turning-point rates sit *below* the reference's, high spatial bands carry near-zero excess energy, and the crossing profile lies *under* the physical budget at every level (Figure 1, left). Raw-coordinate inputs reproduce the same signature (0.879), as does the full 100k-iteration protocol [TODO: verify when landed]. On the BF-unstable testbed the vanilla PINN *diverges* in relative $L_2$ ($0.63 \to 1.31$ over training)—but by progressive decorrelation toward an over-smoothed field: its crossing density (0.046 per unit at mid levels) is a third of the reference's (0.132), i.e. the network has *fewer* waves than the physics, while its amplitude stays in $[0.75, 1.2]$. In no run, on no testbed, at no depth, did any field exceed its crossing budget.

### 5.3 Consequences: budget and dampers are inert

Because every failure is under-oscillatory, the one-sided budget is inactive by construction—its loss is zero on these fields, which our runs confirm empirically: budget-augmented training is statistically indistinguishable from vanilla on the front benchmark [TODO: exp4 numbers: budget_ref/phys/scalar vs vanilla, 3 seeds] and on the chaotic testbed [TODO: chaos budget_ref number]. Uniform gradient damping, the natural null control, is likewise inert or mildly harmful [TODO: grad_damp numbers]—there is no excess gradient energy to damp, only missing structure. This is the sober, useful reading: *loss-level oscillation suppressors of any flavor cannot help a PINN whose failure is under-resolution*; the remedies that address propagation failure (causal weighting, curricula, architecture) live on a different axis.

### 5.4 On the discrepancy with the published failure mode

Our reimplementation follows ASPEN's published configuration in equation, parameters, domain, conditions, architecture, sampling, loss weights, optimizer, schedule, and depth; we additionally varied input normalization and doubled the testbed with a BF-unstable regime. We could not elicit growing high-frequency oscillation in any of these runs. We do not conclude that the published observation is wrong—initialization schemes, precision, framework defaults, or unreported details can all select among PINN failure branches. We conclude that the *prevalence* of the oscillatory branch is implementation-sensitive, which matters for how the community motivates oscillation-suppressing machinery: a stabilizer aimed at a branch that a given stack never visits will measure as inert, and the crossing-profile diagnostic of Section 3 detects this mismatch *before* any stabilizer is trained.

## 6 Discussion and limitations

**What survives.** The budget mechanism itself is validated: one-sided, provably silent under budget, and effective in vitro where excess crossings exist. The crossing-profile diagnostic is, to our knowledge, new to PINN failure analysis and costs one forward/backward pass. The negative result is bounded and specific: 1D CGLE, front and phase-turbulent regimes, the published protocol family, our stack.

**What we did not test.** Deep defect chaos ($|A| \to 0$ defects), other stiff families (Kuramoto–Sivashinsky, reaction–diffusion with sharp fronts), 2D/3D, and PINN variants whose training dynamics differ enough to visit the oscillatory branch (e.g., very high-frequency Fourier features with aggressive learning rates—the regime where our in-vitro patient lives). If a stack does exhibit budget-violating fields, Section 4 is direct evidence the clamp then works; finding a PDE setting where that occurs organically is the open question this paper hands to the community, receipts included.

## 7 Conclusion

We built the loss that the "catastrophic high-frequency divergence" narrative calls for, proved it behaves as specified in both activation and silence, and then could not find the disease it cures in the very setting that motivates it. Both halves are the contribution: a validated one-sided crossing budget—the cap direction of a mechanism whose demand direction helps implicit neural representations—and a documented, reproducible absence, with a one-plot diagnostic that tells any practitioner which failure branch their PINN is actually on before they reach for a stabilizer.

**Reproducibility.** Code, both reference solvers, all raw run JSONs, and the unit tests are public at `https://github.com/gunnerhowe/Research` (directory `kac-rice`; Phase-2 modules `cgle.py`, `pinn.py`, `crossing.py::CrossingBu` runner `experiments/run_phase2.py`). Single consumer workstation (RTX 3080).

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# A Hyperparameters

**Table 1:** Phase-2 hyperparameters.

| | |
|---|---|
| Budget loss | $L$=16 levels on $[-1.05, 1.05]$; $\varepsilon$=0.08; weight 1.0 |
| Budget sources | ref. sim (q95 ×1.5 slack); physics prior $m$=4; scalar cap |
| Vanilla PINN | $8 \times 40$ tanh; Adam $10^{-3} \to 10^{-4}$ at half-depth |
| Collocation | 20,000 res. + 1,000 IC + 1,000 BC, Latin hypercube |
| Chaotic testbed | $b$=2, $c$= − 1.2, $L$=64, $T$=5, periodic embedding, 4 harmonics |
| In vitro | SIREN $\omega_0$=30, 40 samples, budget $m$=6 over $[-1, 1]$ |
| Seeds | 3 per configuration |

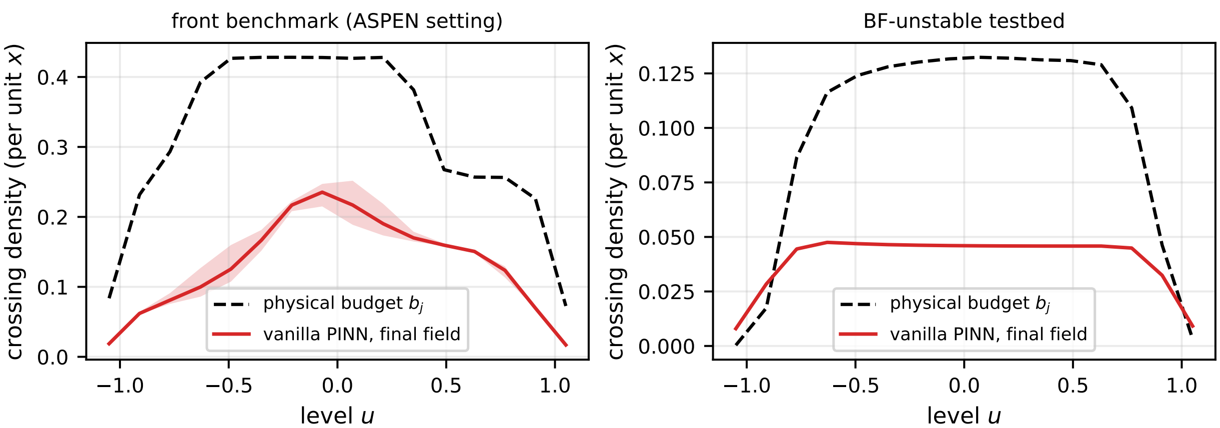


**Figure 1: The diagnostic in action**: measured crossing profiles versus physical budgets for the front benchmark (left) and the BF-unstable testbed (right). Every failed vanilla field sits *below* budget—the under-oscillatory branch, on which one-sided budgets are by design inert. [TODO: generate figure]

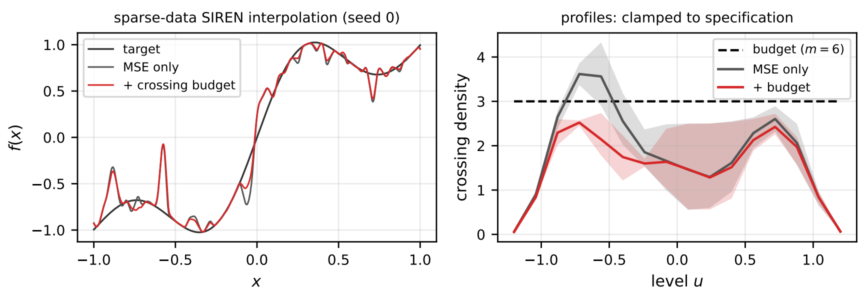


**Figure 2: In-vitro positive control**: sparse-data SIREN interpolation. MSE-only over-oscillates beyond budget; the one-sided budget clamps the profile within specification and improves test error on all seeds. [TODO: generate figure]